

temperature- and relative humidity-regulatory devices (such as compressors and thermostats) in place. Desired temperature range for a cold storage is 5°C to 25°C. Relative humidity should be between 65 and 95 per cent. The room itself must be made of various insulating materials like polycarbonates for enhanced thermal regulation.

Solar panels collect solar energy and power the compressors and other appliances of the storage room. Compressors can be adjusted as per cooling requirements—shelf-wise or at room-level. The system can be made grid-connected or completely off-grid, as suitable. Excess energy can be stored either in batteries or thermal storage units.

Cost points

High investment remains a major obstacle for such solutions to find large-scale acceptance. Bheke gives an idea, “A storage room setup of about three- to four-tonne capacity, including electric appliances and thermal storage units, costs about ₹ 400,000 to ₹ 500,000, while solar panels draw a separate cost. Tentative cost for the panels is ₹ 100,000 per kW capacity. The final price is calculated as per the panels and storage capacities required.” The overall cost is 30 to 50 per cent higher as compared to traditional cold storages. Lack of domestic production of solar hardware units is the major reason for this high cost.

The best way to invest is in groups. Bheke continues, “It may be difficult for an individual mid-scale farmer to invest in a store room. It is easier for a farmer union or a group to gather funds



Solar panels can run cold storage units using green energy and bring great benefits (Credit: flickr.com)



A representation of a solar cold storage setup (Credit: vertcaptech.com)

and invest in a store room for their common use.”

If batteries are used, these need to be changed every four to five years, which further adds to the operational cost. Altogether, the investment needs to be worked out.

Government subsidies

The government of India is providing subsidies under various schemes for solar storage investment. Ministry of New and Renewable Energy (MNRE) has extended subsidies for the same. For instance, a document by MNRE proposes that a plan for solar cold storage having a capacity of five to six tonnes and 4kWp solar panels,

which can cost ₹ 900,000, will be eligible for a subsidy amount of ₹ 585,000.

The administration is actively promoting solar-driven applications in rural areas through such perks. Yet, some challenges exist that need to be addressed. “Subsidies are available, but most schemes require the farmers to pay full at first and then get the return based on application,” says Bheke.

Case studies

India is going strong with many solar cold storage installations in recent years. For example, MNRE shared a recent project at Bhopal, where an eight-tonne capacity solar storage with 20kWp solar panels and lead-acid batteries was installed. The complete installation had cost ₹ 2 million. The installation cut down operational expenses by 28 per cent, while saving 16,500kWh electricity and ₹ 115,000 per year. It improved crop shelf-life by up to 21 days. While the calculated ROI period is about nine years, the benefits have already started to come in with consistent payback.

Etawah, Uttar Pradesh, also saw its first successful launch of solar storage in 2015. Installed by Statcon Energiaa for Shree Cold Storage, the solution replaced the existing electricity-driven setup. It saves 700 units of electricity per day, culminating into ₹ 150,000 per month.

With all the initiatives in motion, a little more push by the government and higher participation from industry players can transform solar cold storages into a big boon for India. **EFY**

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